Flexible ranking with biased, higher-order random walk simulation

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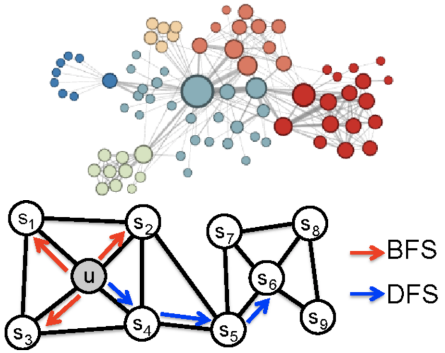
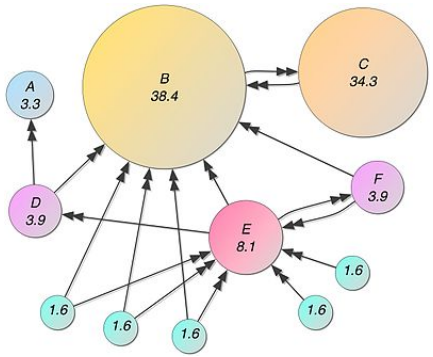
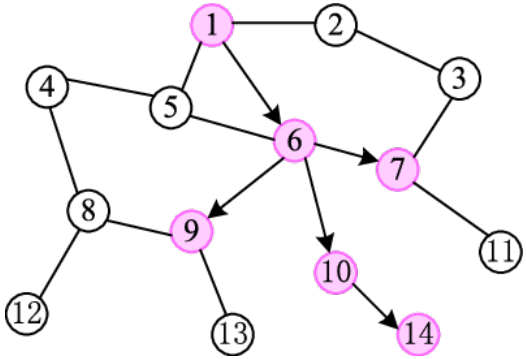
Random walks and graph analytics



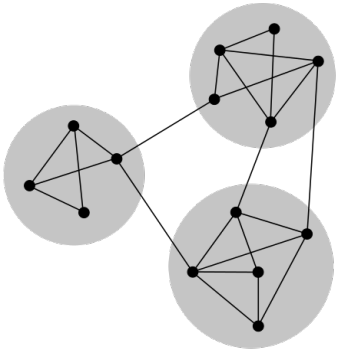
(a) Embedding (e.g. node2vec)
Grover, Leskovec, "node2vec: Scalable Feature Learning for Networks." (2016).



(b) Ranking (e.g. PageRank)
Page, et al. "The PageRank Citation Ranking: Bringing Order to the Web." (1999).



(c) Modeling diffusion processes
Zhang, et al. "Dynamics of information diffusion and its applications on complex networks." (2016).



(d) Community detection (e.g. WalkTrap)
Pons, Lapaty, "Computing communities in large networks using random walks." (2005).

Figure: Applications of random walks in graph analytics

Ranking and non-Markovian data

PageRank scores are determined using simulated *simple random walks* on the graph; however, this first-order (memoryless) assumption is oftentimes unrealistic.

Figure: Non-Markovian shipping data
Xu, et al. "Representing higher-order dependencies in networks." (2016).

Simulating biased random walks at scale

We have an asynchronous HPC graph engine **HavoqGT**[3] that is well-suited for biased random walk simulation.

- Developed under family of LLNL Projects (FY13-19)
- C/C++ with MPI, custom message routing/aggregation
- Support for scale-free graphs
- HavoqGT good at graph traversal, several Graph500[4] successes

Massive state-spaces involved $\implies$ need for simulation convergence theory

[3] Pearce, et al., HavoqGT, https://github.com/LLNL/havoqgt
[4] Murphy, et al., "Introducing the Graph 500" (2010)

Our contributions

- Develop a **flexible ranking** framework which uses **biased random walks**.
- Extend the existing **convergence theory**[5] for ranking based on simple random walks.
- Present a **scalable** way to do higher-order **complex network analysis**.

[5] Avrachenkov, et al. "Monte Carlo Methods for Top-k Personalized PageRank Lists and Name Disambiguation." (2010).

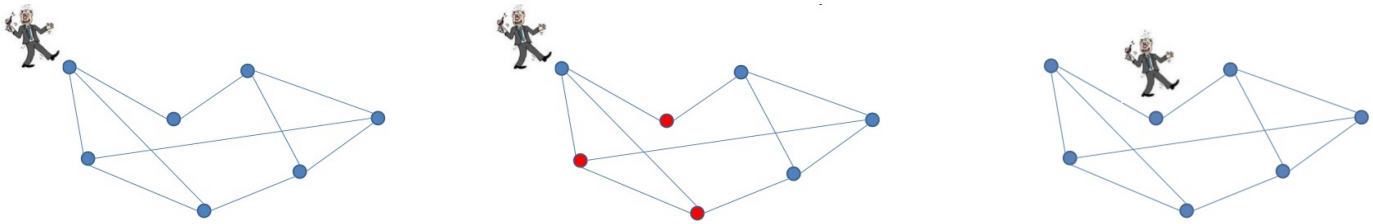
Outline

1. Background
   - Simple random walks
   - Personalized PageRank
   - Existing convergence theory
2. Ranking with higher-order random walks
   - Motivation
   - Algorithms and convergence theory
   - Temporally-biased walker experiment
3. Conclusions and future work

Simple random walks

A simple random walk on graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$

- Sequence of nodes $\{v_0, v_1, v_2, \cdots\}$ where v_{t+1} is chosen uniformly at random from the neighbors of v_t



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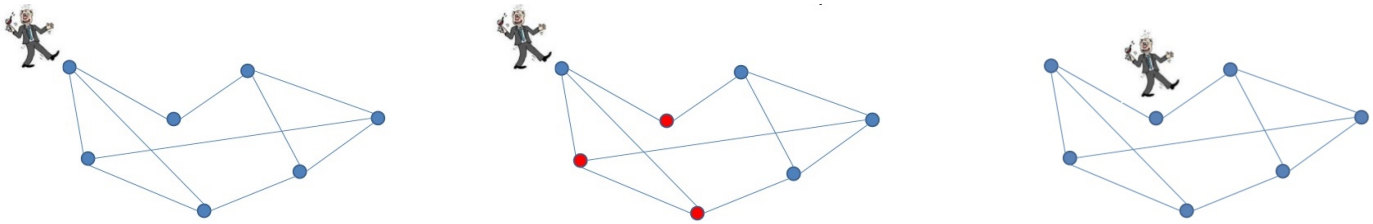
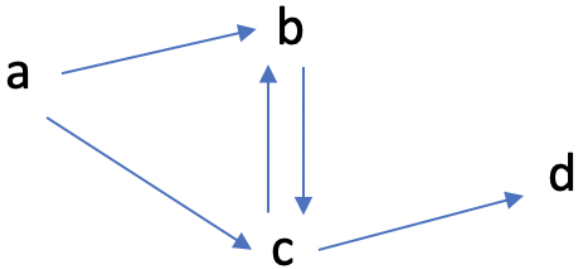


- Represent with row-stochastic **transition matrix** P, where

$$P_{ij} = \mathbb{P}[v_{t+1} = j | v_t = i] = \mathbb{P}[\text{transition from } i \text{ to } j]$$



Graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$

$$\begin{bmatrix} 0 & 1/2 & 1/2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1/2 & 0 & 1/2 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Transition matrix P

Top image taken from: https://www.slideshare.net/pavankapanipathi/random-walk-on-graphs

Personalized PageRank[6]

Slight modification:

- With probability α, transition to random out-neighbor
- With probability $1 - \alpha$, jump back to seed node

$$\alpha \begin{bmatrix} 0 & {}^1/_2 & {}^1/_2 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & {}^1/_2 & 0 & {}^1/_2 \\ 1 & 0 & 0 & 0 \end{bmatrix} + (1-\alpha) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

[6] Haveliwala. "Topic-sensitive PageRank." (2002)

Personalized PageRank[6]

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Irreducible + aperiodic $\implies$ **unique stationary distribution** exists

$$\boldsymbol{\pi} = \boldsymbol{\pi}\left(\alpha P + (1-\alpha)\mathbb{1}^T \boldsymbol{e_s}\right)$$

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Irreducible + aperiodic $\implies$ **unique stationary distribution** exists

$$\boldsymbol{\pi} = \boldsymbol{\pi}\left(\alpha P + (1-\alpha)\mathbb{1}^T \boldsymbol{e_s}\right)$$

$$\boldsymbol{\pi} = \begin{bmatrix} 0.27 & 0.256 & 0.332 & 0.141 \end{bmatrix}$$

[6] Haveliwala. "Topic-sensitive PageRank." (2002)

Personalized PageRank

Avrachenkov, et al.[7] provide two ways of estimating $\boldsymbol{\pi}$ by simulating the **Personalized PageRank random walk**:

Start walker at seed node s.
With probability α, transition to a random out-neighbor.
With probability $1 - \alpha$, terminate.

[7] Avrachenkov, et al. "Monte Carlo Methods for Top-k Personalized PageRank Lists and Name Disambiguation." (2010).

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EP (end point) interpretation:

$$\boldsymbol{\pi}_i = \mathbb{P}[\text{terminate at } i]$$

CP (complete path) interpretation:

$$\boldsymbol{\pi}_i = (1-\alpha)\mathbb{E}[\#\text{visits to } i]$$

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EP (end point) interpretation

$$\pi_i = \mathbb{P}[\text{terminate at } i]$$

Algorithm (EP): Simulate m walks starting at node s. Evaluate π_i as the fraction of those terminating at node i.

Convergence analysis:

$$\mathbb{E}\Big[\hat{\pi}_i(m)\Big] = \pi_i$$

$$\mathrm{Var}\Big[\hat{\pi}_i(m)\Big] = \frac{1}{m}\pi_i\Big(1 - \pi_i\Big)$$

CP (complete path) interpretation

$$\boldsymbol{\pi}_i = (1-\alpha)\mathbb{E}[\#\text{visits to } i]$$

Algorithm (CP): Simulate m walks starting at node s. Evaluate $\boldsymbol{\pi}_i$ as the total number of visits to node i across all walks, multiplied by $\frac{1-\alpha}{m}$.

Convergence analysis:

$$\mathbb{E}\left[\hat{\hat{\boldsymbol{\pi}}}_i(m)\right] = \boldsymbol{\pi}_i$$

$$\mathrm{Var}\left[\hat{\hat{\boldsymbol{\pi}}}_i(m)\right] \approx \frac{1}{m}\boldsymbol{\pi}_i\Big(1-\alpha-\boldsymbol{\pi}_i\Big)$$

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Higher-order random walks

In a **higher-order random walk**, transition probabilities can be biased by previously visited nodes and graph metadata.

Higher-order random walks

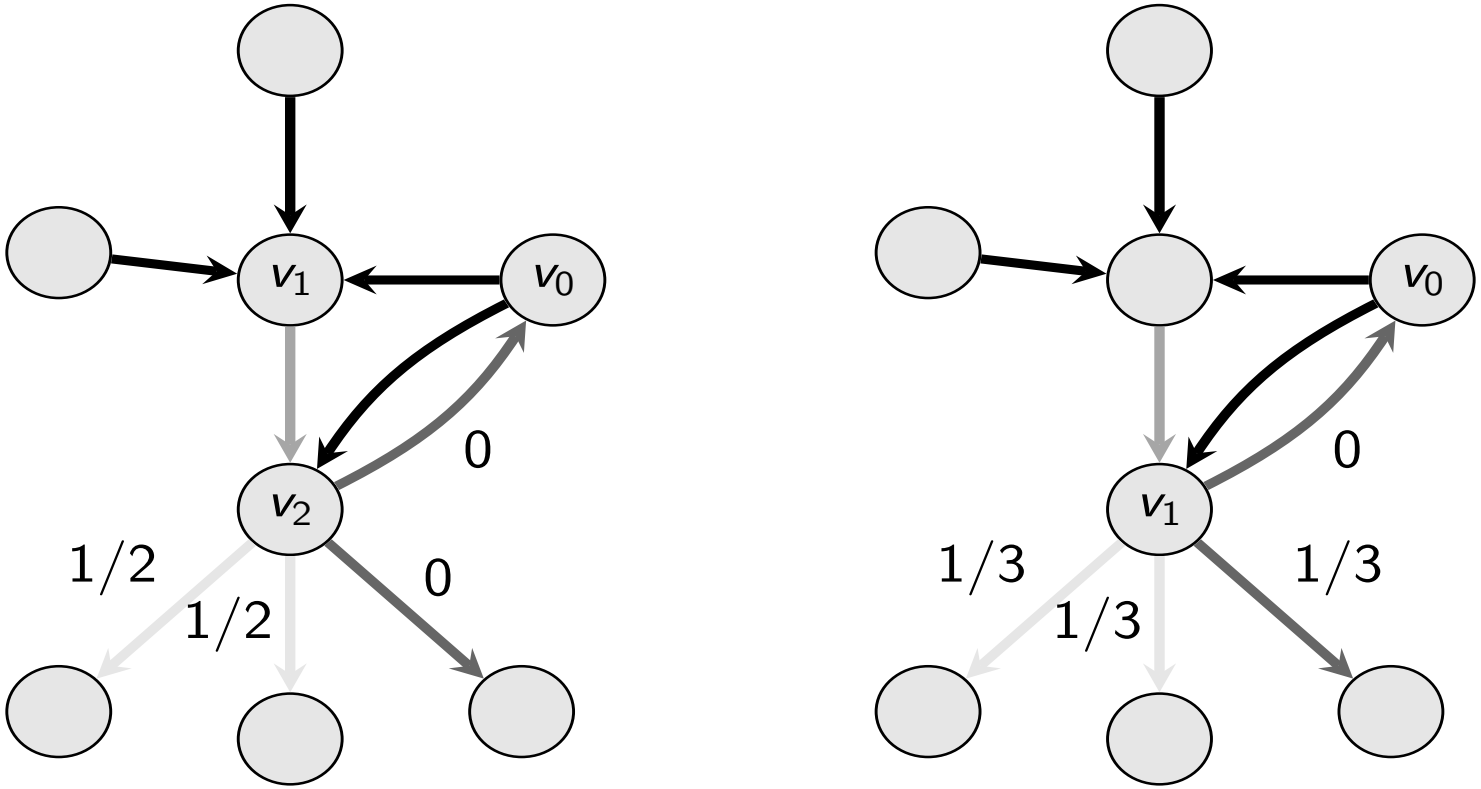


Figure: Temporally-causal (time flows dark to light)

Higher-order random walks

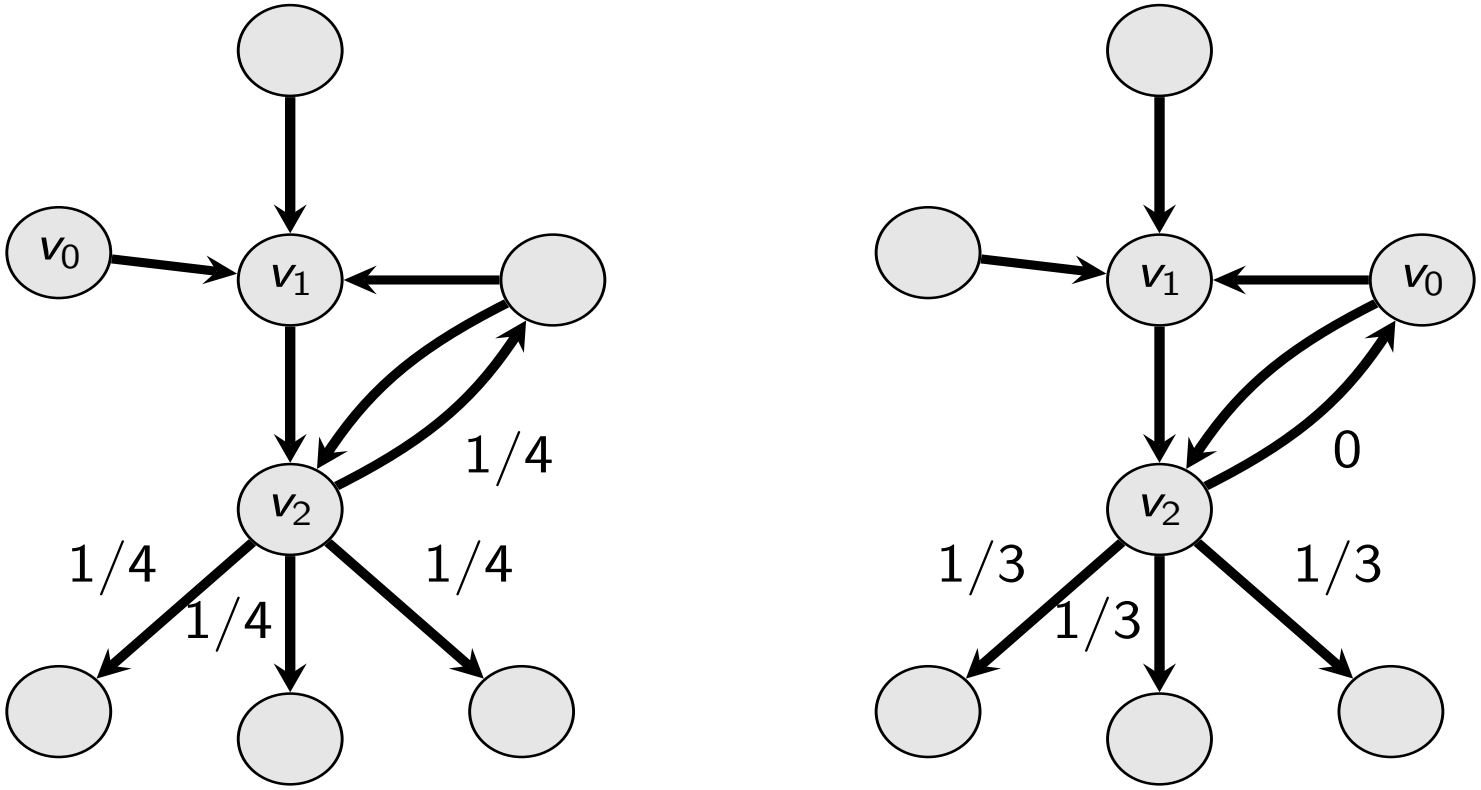


Figure: Self-avoiding

Higher-order random walks

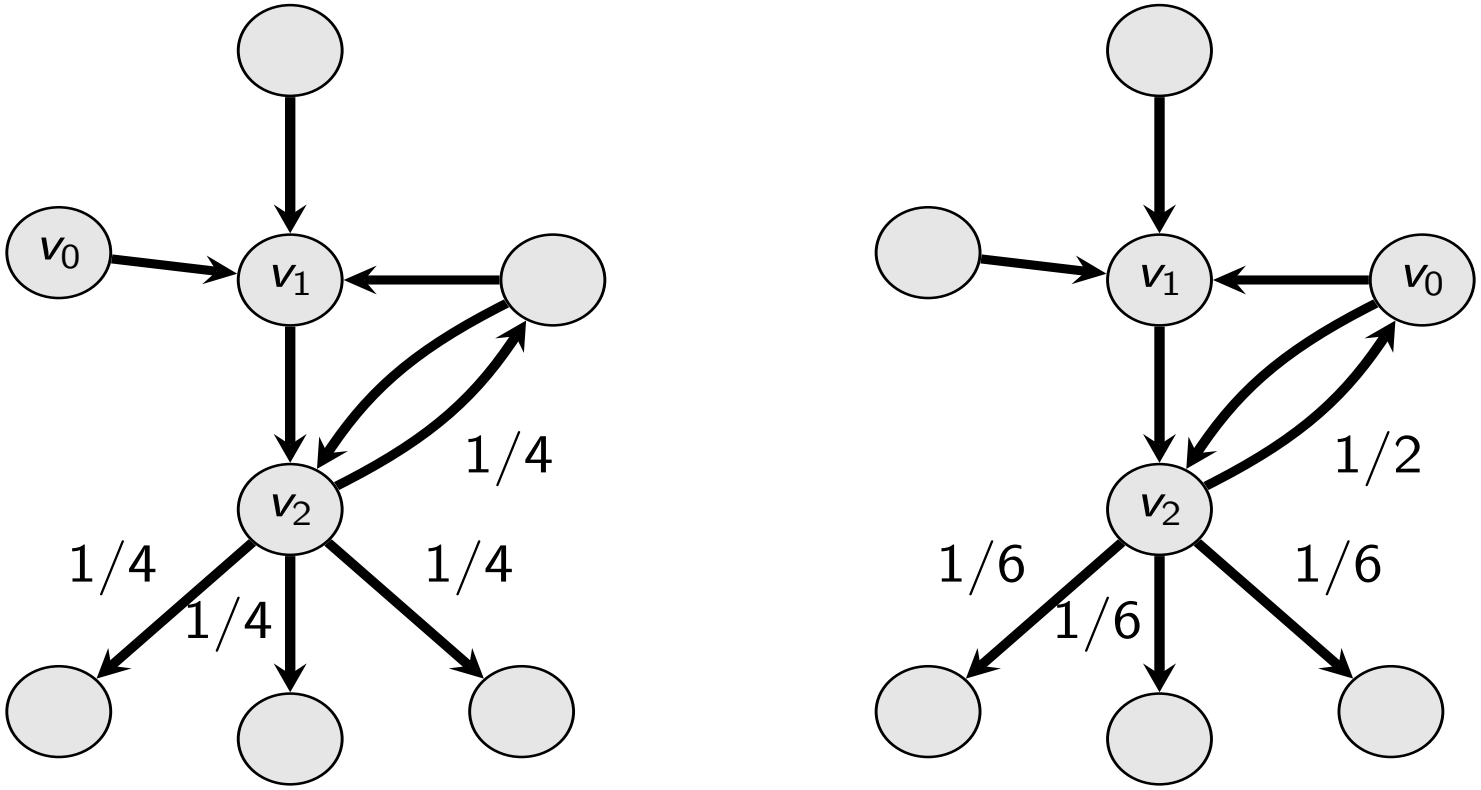
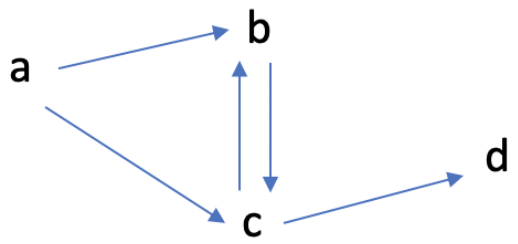
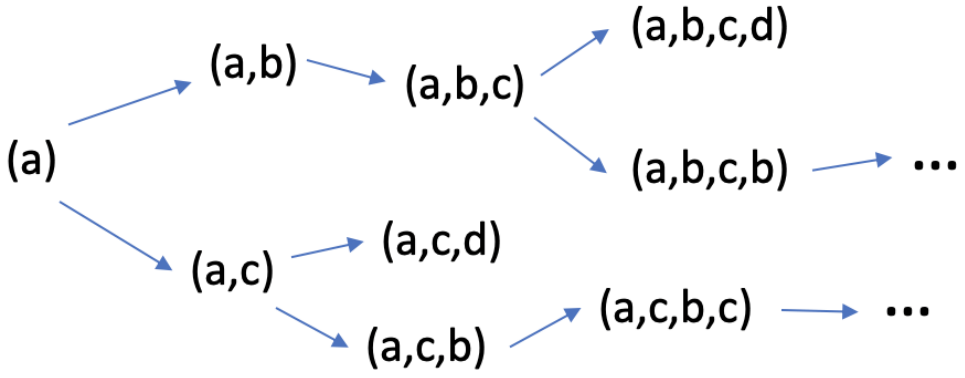


Figure: Triangle-biased (50%)

Higher-order random walks

How can we represent with a transition matrix?





Markovian on higher-dimensional state space

- $\mathcal{O}\Big(\sum_{\ell=1}^{h}$ paths of length $\ell\Big)$, where $h =$ max history length
- states are **ordered tuples of nodes**
- transitions are **model-dependent**
- call transition matrix P'

Higher-order random walks

Embedded simple random walk, $h = 3$, seed $= (a)$

	a	b	c	d	ab	ac	bc	cb	cd	abc	acb	acd	bcb	bcd	cbc
a					1/2	1/2									
b							1								
c								1/2	1/2						
d	1														
ab										1					
ac											1/2	1/2			
bc													1/2	1/2	
cb															1
cd	1														
abc													1/2	1/2	
acb															1
acd	1														
bcb															1
bcd	1														
cbc													1/2	1/2	

Higher-order random walks

Self-avoiding, $h = 3$, seed $= (a)$

	a	b	c	d	ab	ac	bc	cb	cd	abc	acb	acd	bcb	bcd	cbc
a					1/2	1/2									
b							1								
c								1/2	1/2						
d	1														
ab										1					
ac											1/2	1/2			
bc													0	1	
cb	1														0
cd	1														
abc													0	1	
acb	1														0
acd	1														
bcb	1														0
bcd	1														
cbc													0	1	

Ranking with higher-order random walks

Solve for the stationary distribution and we're done?

$$\boldsymbol{\pi}' = \boldsymbol{\pi}' \left(\alpha P' + (1-\alpha) \mathbb{1}^T \boldsymbol{e}_{(s)} \right)$$

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Definition (Higher-order ranking of node i)

$$\boldsymbol{\theta}_i := \sum_{\psi \in \Psi_i} \boldsymbol{\pi}'_\psi$$

where Ψ_i is the set of all states "ending" in i: $(i), (\diamond, i), (\diamond, \diamond, i), \cdots$

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Can similarly estimate by simulating random walks on this space.

Ranking with higher-order random walks

Higher-order Personalized PageRank random walk (HOPPRW)

Pseudocode

```
state ← (s)
while(continueWalking[α]):
    D ← {neighbors[state[−1]] which satisfy constraints[state]}
    if(isEmpty[D]):
        state ← (s)
    else:
        nextNode ← sampleFrom[D]
        state ← append[state, nextNode]
```

Ranking with higher-order random walks

EP (end point) interpretation

Higher-order EP interpretation

$$\begin{aligned}\boldsymbol{\theta}_i &= \sum_{\psi \in \boldsymbol{\Psi}_i} \boldsymbol{\pi}'_\psi \\ &= \sum_{\psi \in \boldsymbol{\Psi}_i} \mathbb{P}[\text{terminates at } \psi] \\ &= \mathbb{P}\Big[\bigcup_{\psi \in \boldsymbol{\Psi}_i} \text{terminates at } \psi\Big]\end{aligned}$$

Algorithm (Higher-order EP): Simulate m walks starting at state (s). Evaluate $\boldsymbol{\theta}_i$ as the fraction of those terminating at any state in $\boldsymbol{\Psi}_i$.

Ranking with higher-order random walks

EP (end point) interpretation

Theorem (Higher-order EP)

$$\mathbb{E}\left[\hat{\boldsymbol{\theta}}_i(m)\right] = \boldsymbol{\theta}_i$$

$$\mathrm{Var}\left[\hat{\boldsymbol{\theta}}_i(m)\right] = \frac{1}{m}\boldsymbol{\theta}_i\left(1 - \boldsymbol{\theta}_i\right)$$

Ranking with higher-order random walks

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Corollary

Number of walkers m needed to estimate $\boldsymbol{\theta}_i$ with relative error ϵ at $100(1-\delta)\%$ confidence is

$$m = \frac{z_{\delta/2}^2}{\epsilon^2}\frac{1 - \boldsymbol{\theta}_i}{\boldsymbol{\theta}_i}$$

Ranking with higher-order random walks

CP (complete path) interpretation

Higher-order CP interpretation

$$\begin{aligned}\boldsymbol{\theta}_i &= \sum_{\psi \in \boldsymbol{\Psi}_i} \boldsymbol{\pi}'_\psi \\ &= \sum_{\psi \in \boldsymbol{\Psi}_i} (1-\alpha)\mathbb{E}[\#\text{visits to } \psi] \\ &= (1-\alpha)\mathbb{E}\Big[\sum_{\psi \in \boldsymbol{\Psi}_i} \#\text{visits to } \psi\Big]\end{aligned}$$

Algorithm (Higher-order CP): Simulate m walks starting at state (s). Evaluate $\boldsymbol{\theta}_i$ as the total number of visits to any state in $\boldsymbol{\Psi}_i$ across all walks, multiplied by $\frac{1-\alpha}{m}$.

Ranking with higher-order random walks

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Theorem (Higher-order CP)

$$\mathbb{E}\left[\hat{\hat{\boldsymbol{\theta}}}_i(m)\right] = \boldsymbol{\theta}_i$$

$$\mathrm{Var}\left[\hat{\hat{\boldsymbol{\theta}}}_i(m)\right] \leq \frac{1}{m}\boldsymbol{\theta}_i\left(\frac{1+\alpha^2}{1+\alpha} - \boldsymbol{\theta}_i\right)$$

Ranking with higher-order random walks

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Corollary

Higher-order CP estimate has less variance than the higher-order EP one.

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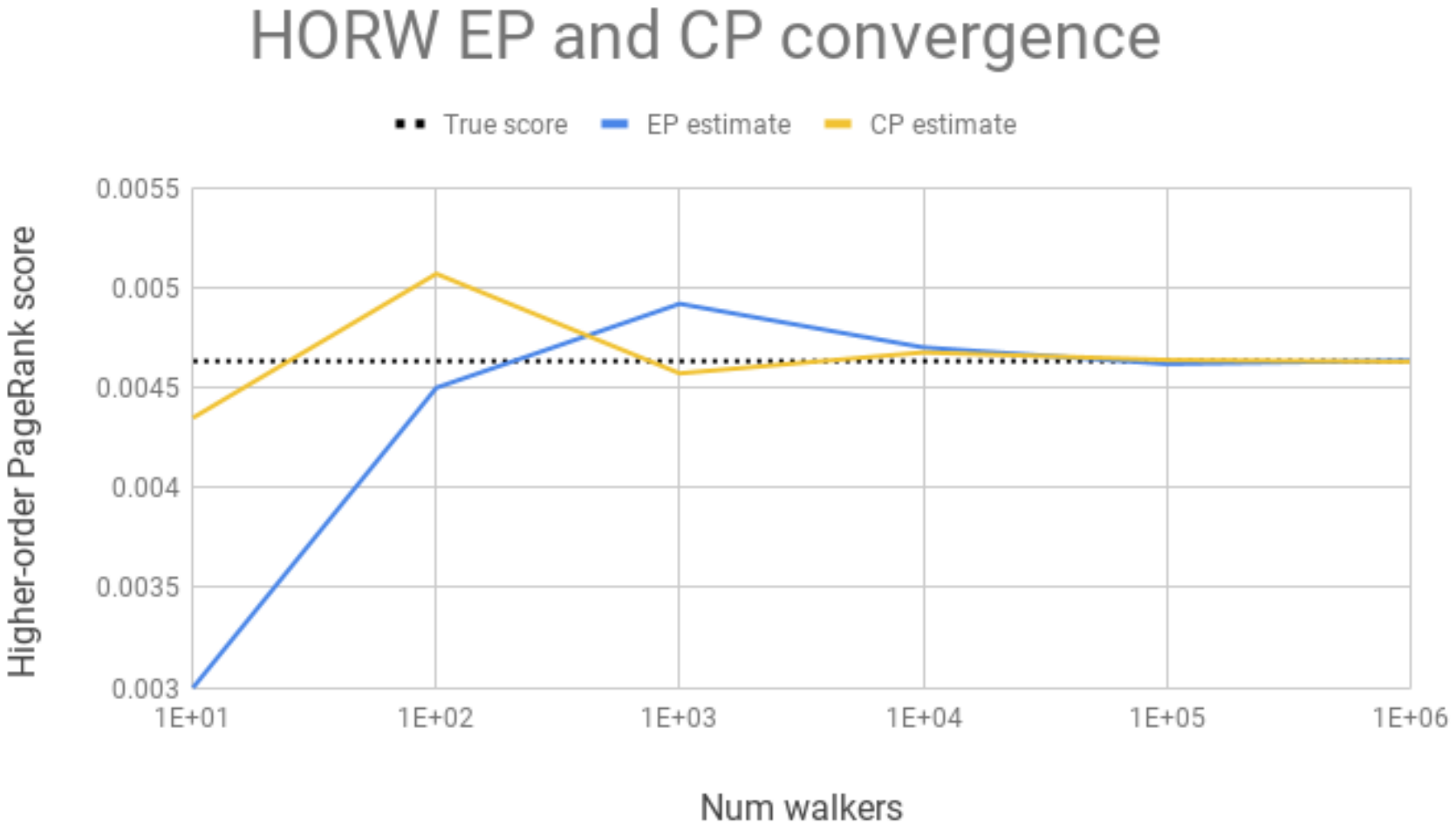
Temporally-biased walk experiment

- SNAP[8] twitter graph of Higgs boson rumor spreading
- Derivative graph (with timestamps) where $i \to j$ when $i \neq j$ and either i mentions j or j re-tweets i
  - $\approx$ 300k vertices
- **Temporally-biased** random walk strategy, measure of "node influence"
- Plotted **estimates averaged over 100 runs** for both EP and CP scores for varying numbers of walkers

[8] Leskovec, Krevl, "SNAP Datasets: Stanford Large Network Dataset Collection." http://snap.stanford.edu/data

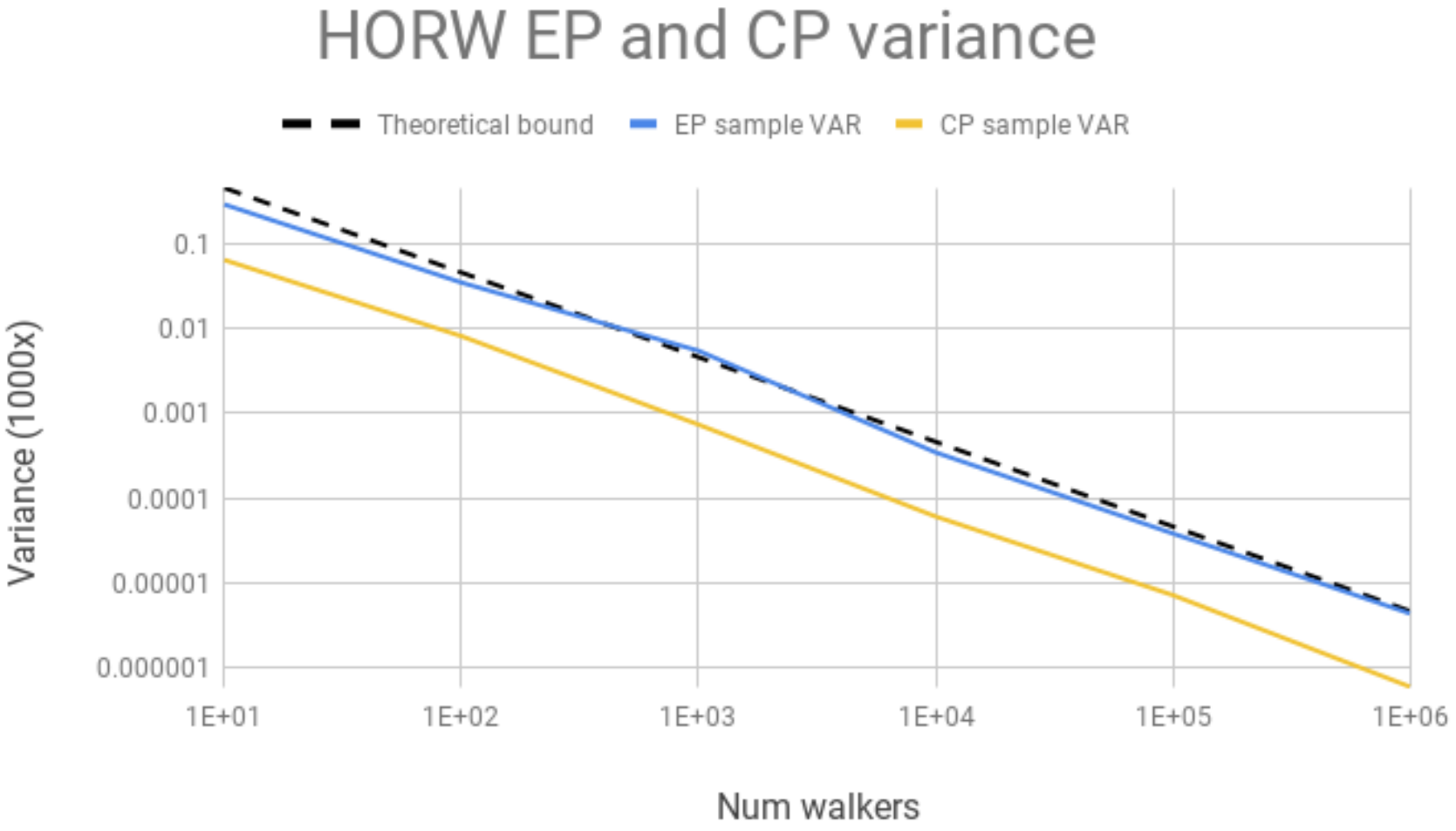
Temporally-biased walk experiment



Seed = 200223, Target = 81450

(ranked 10th by HOPPRW, 22nd by PPRW)

Temporally-biased walk experiment



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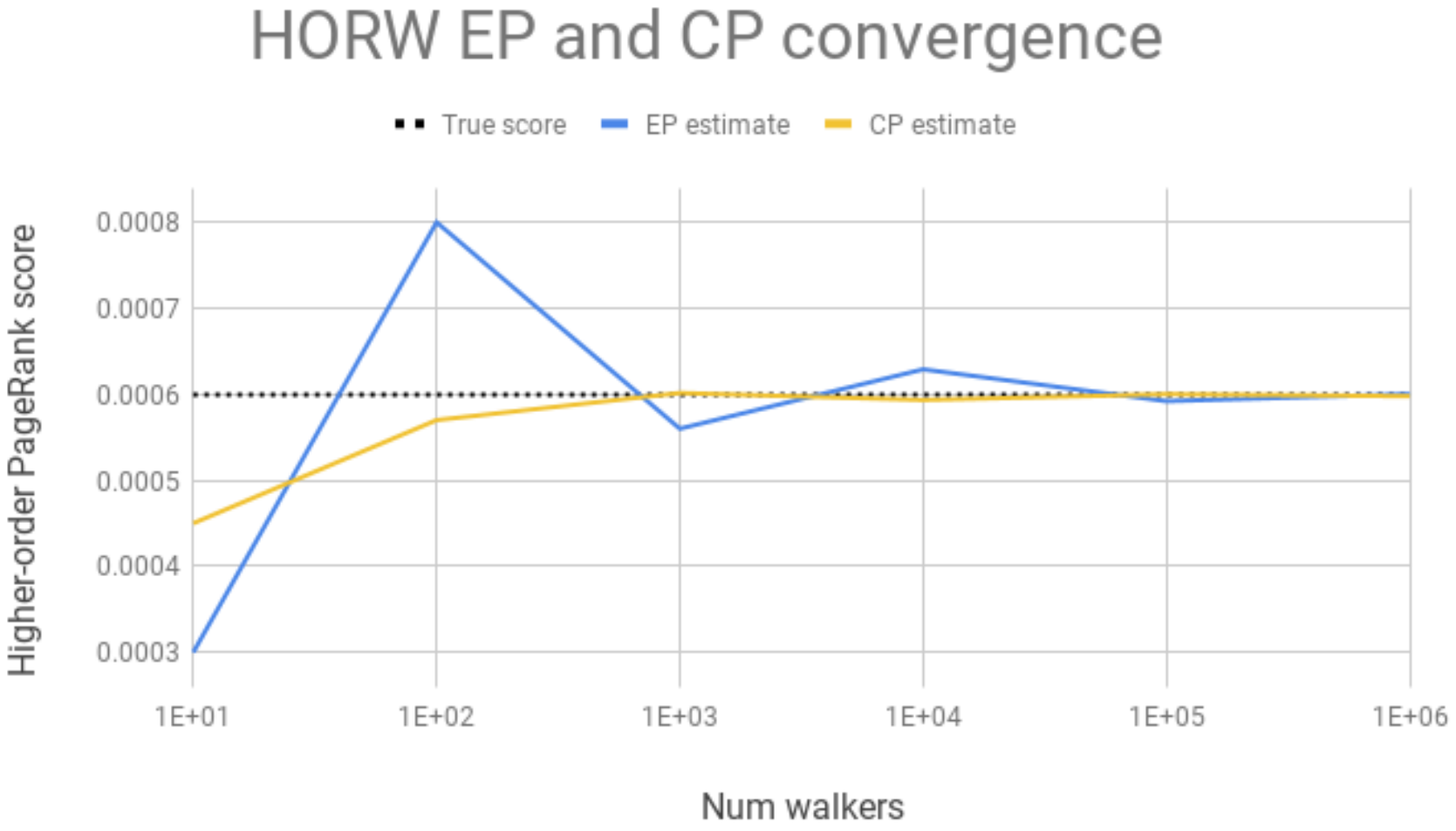
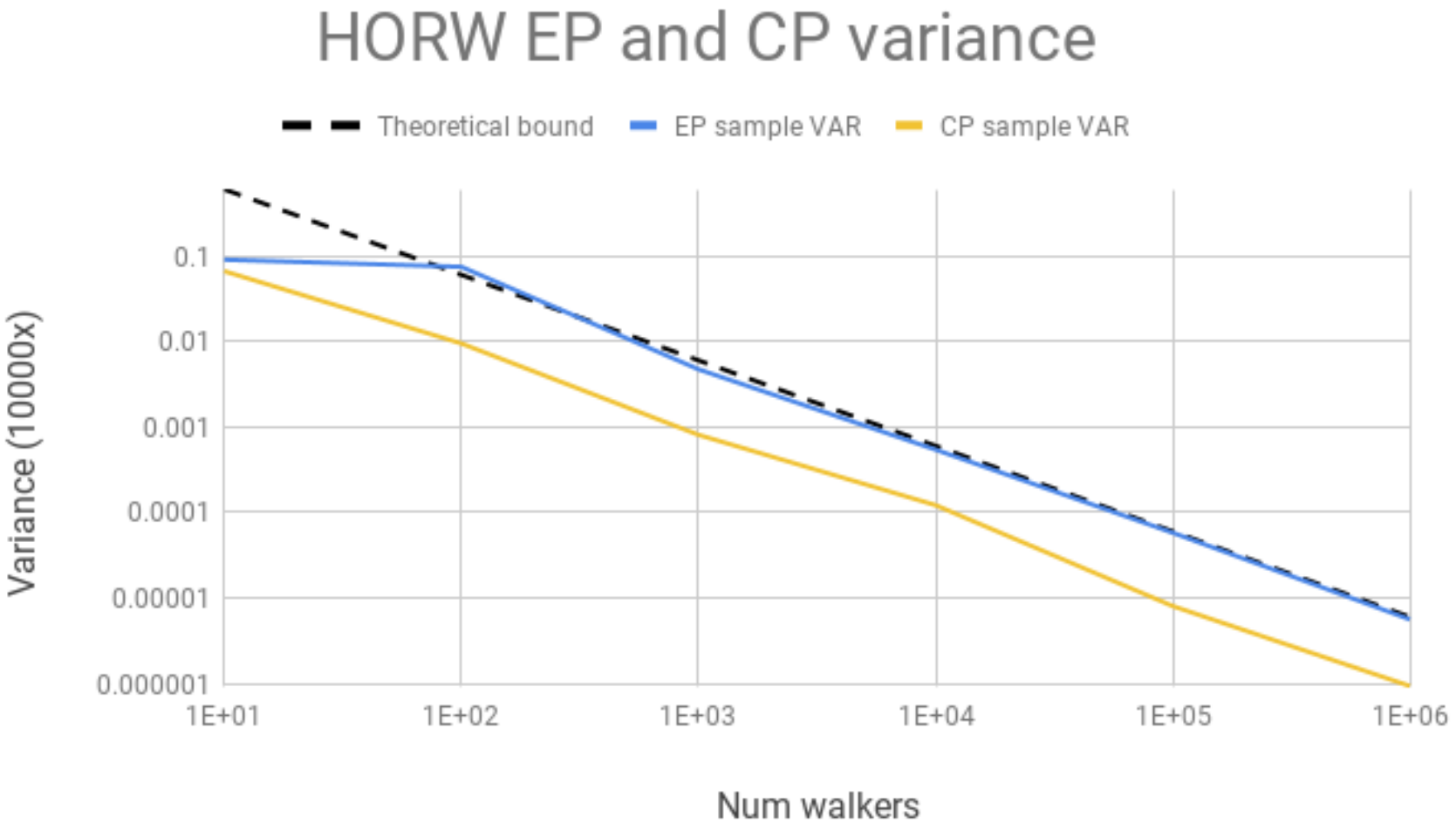


Seed = 200223, Target = 34030

(ranked 100th by HOPPRW, 124th by PPRW)

Temporally-biased walk experiment



Seed = 200223, Target = 34030

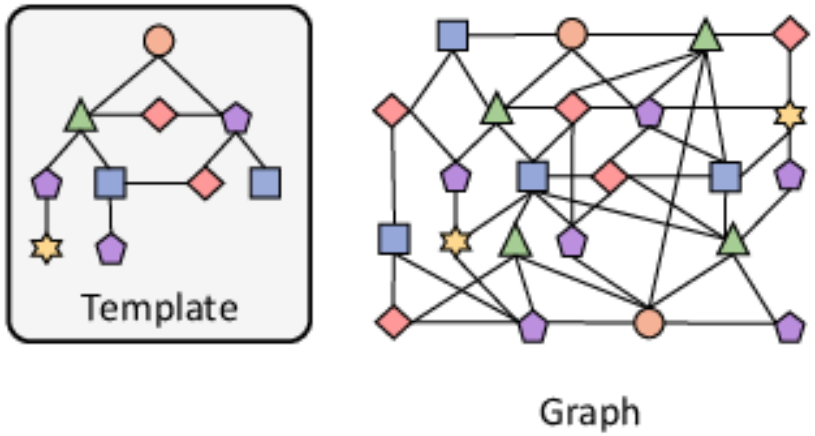
(ranked 100th by HOPPRW, 124th by PPRW)

Summary

- Extended PageRank **convergence theory** to higher-order random walks
- Demonstrated use case on a real-world temporal network; highlighted others
- Presented a **computationally feasible** way to do higher-order **complex network analysis** using random walks

Future work

- Formalize top-k convergence theory and stopping criteria
- *Learning* the constraints and transition rules
- Applications to approximate pattern matching[9]



[9] Reza, et al. "PruneJuice: Pruning Trillion-edge Graphs to a Precise Pattern-Matching Solution." (2018).